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AC-DC FLYBACK CONVERTER FOR MOSFET DRIVING
VOLTAGE REQUIREMENT

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AC-DC FLYBACK CONVERTER
FOR MOSFET DRIVING VOLTAGE REQUIREMENT

A GRADUATION/DESIGN PROJECT
by

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Abstract

This project covers the design and implementation of a 60W Flyback Converter that provides efficient switching and control using a UC3844 PWM control IC. The converter steps down the 220V AC input voltage to a regulated 20V DC output with a maximum current capacity of 3A, making it suitable for low and medium power applications. The transformer used in the converter has a 5:1 turns ratio and a primary inductance of 329 μH and is constructed with an ETD34 ferrite core and enamelled copper wire, providing efficient energy transfer and electrical isolation. The output voltage is regulated by a TL431-based feedback circuit, and optical isolation is provided by an optoisolator to maintain safety and reliability. The proportional-integral (PI) control loop provides precise voltage regulation under different load conditions, while the overvoltage protection (OVP) circuit protects the system and connected loads from abnormal voltage spikes. Thermal management is provided by an aluminium heat sink connected to the output rectifier diode, ensuring efficient heat dissipation and reliable operation under full load. The converter achieves 85% efficiency by reducing energy losses and minimizes reactive power consumption with an 85% power factor, ensuring efficient use of input energy. The compact 13x9.5 cm PCB design makes this converter suitable for space-limited applications in areas such as consumer electronics, industrial control systems, renewable energy systems and portable power supplies. This project highlights the potential of the Flyback topology for sustainable energy conversion, focusing on its high efficiency, reliability and compact design features. Future improvements, such as extending the input voltage range, optimizing the feedback loop, and optimizing the PCB design for a more optimal design could further improve performance and application scope.

Keywords: Flyback, PCB Design, Schematic, PWM, Efficiency

Özet

Bu proje, UC3844 PWM kontrol entegresi kullanarak verimli anahtarlama ve kontrol sağlanan 60W Flyback Dönüştürücüsünün tasarımı ve uygulanmasını kapsamaktadır. Dönüştürücü, 220V AC giriş gerilimini, maksimum 3A akım kapasitesine sahip, düzenlenmiş 20V DC çıkışa düşürmektedir; bu da cihazı düşük ve orta güç uygulamaları için uygun hale getirmektedir. Dönüştürücüde kullanılan trafo, 5:1 sarım oranına ve 329 μ H birincil endüktansa sahip olup, ETD34 ferrit çekirdek ve emaye kaplanmış bakır tel ile inşa edilmiştir; böylece verimli enerji aktarımı ve elektriksel izolasyon sağlanmaktadır. Çıkış voltajı, TL431 tabanlı geri besleme devresiyle düzenlenmekte ve bir optoizolatör aracılığıyla optik izolasyon sağlanarak güvenlik ve güvenilirlik korunmaktadır. Proportional-Integral (PI) kontrol döngüsü, farklı yük koşullarında hassas voltaj düzenlemesi sağlamakta, aşırı voltaj koruma (OVP) devresi ise sistem ve bağlı yüklerin anormal gerilim yükselmelerinden korunmasına olanak tanımaktadır. Termal yönetim, çıkış doğrultma diyoduna bağlanan alüminyum ısı emici ile sağlanarak, ısıнын verimli bir şekilde dağıtılması ve tam yük altında güvenilir işletim temin edilmektedir. Dönüştürücü, enerji kayıplarını azaltarak %85 verimlilik elde etmekte ve %85 güç faktörü ile reaktif güç tüketimini minimize ederek giriş enerjisinin verimli kullanımını sağlamaktadır. Kompakt 13x9,5 cm PCB tasarımı, bu dönüştürücüyü, tüketici elektroniği, endüstriyel kontrol sistemleri, yenilenebilir enerji sistemleri ve taşınabilir güç kaynakları gibi alanlarda, sınırlı alana sahip uygulamalar için uygun hale getirmektedir. Bu proje, Flyback topolojisinin sürdürülebilir enerji dönüşümü potansiyelini vurgulamakta, yüksek verimlilik, güvenilirlik ve kompakt tasarım özelliklerine odaklanmaktadır. Gelecekteki iyileştirmeler, örneğin giriş gerilim aralığının genişletilmesi, geri besleme döngüsünün optimize edilmesi, PCP tasarımında daha optimum tasarım performans ve uygulama kapsamını daha da geliştirebilir.

Anahtar Kelimeler: Flyback, PCB Tasarımı, Şema, PWM, Verimlilik

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ACRONYMS

AC Alternating Current

DC Direct Current

EMI Electromagnetic Interference

PWM Pulse Width Modulation

ZCS Zero Current Switching

ZVS Zero Voltage Switching

CMC Common Mode Choke

MOSFET Metal Oxide Semiconductor Field Effect Transistor

SMPS Switch Mode Power Supply

IC Integrated Circuit

OPAMP Operational Amplifier

I. INTRODUCTION

Reference to subsection 1.1., 1.2. Reference to Table 1-1, 1-2.

1.1. Constraints of the Design Project

As shown in Table 1-1, the designed project was evaluated according to the appropriate constraints.

Table I- 1 Constraints of the Design Project

No	Constraint Name	Description of the Constraint
1	Economical	The project prioritized the use of low-cost, easily available components to ensure affordability. A detailed cost analysis was conducted, with local suppliers preferred to reduce customs and logistics expenses. The total prototype and PCB production costs were kept within reasonable limits.
2	Environmental	-
3	Sustainability	-
4	Manufacturability	The PCB and circuit design were prepared considering ease of production. A double-layer PCB layout was used with all components placed strategically for efficient assembly in both manual and automated manufacturing environments. Transformer winding details and component specifications were fully documented for reproducibility.
5	Ethical	-
6	Health	-
7	Safety	Key safety measures include galvanic isolation between input and output via a transformer, optocoupler-based feedback for control isolation, input surge protection (varistor, NTC thermistor), snubber circuits for voltage spikes, and Over Voltage Protection (OVP) to prevent overvoltage damage.
8	Social	-
9	Legal	-
10	Political	-
11	Reproducibility	The project has been documented in detail to enable replication. Transformer winding parameters, PCB layout files, and component lists have been shared to ensure reproducibility. Additionally, test procedures and measurement results were recorded, making it possible for others to reproduce the design and verify its performance under similar conditions.
12	Cost and Budget	A strict budget constraint was applied throughout the design and implementation process. Total material, PCB, and prototyping costs were documented in a detailed budget analysis, ensuring the project remained financially feasible.
13	Other	-

1.2. Aims and Objectives

The flyback converter is a widely preferred topology in low to medium power applications due to its simplicity, cost-effectiveness, and ability to provide robust electrical isolation between input and output. This project focuses on the design and implementation of a flyback converter that converts a 220V AC input into a regulated 20V DC output, suitable for driving medium-power loads with reliable performance. At the core of the design, the UC3844 PWM control IC enables high-frequency switching of a MOSFET, facilitating efficient energy transfer through a ferrite core transformer.

The transformer, designed with a 5:1 turns ratio, ensures effective energy transfer while maintaining necessary electrical isolation between the input and output sides. An optically isolated TL431-based feedback system is employed for voltage regulation, enhancing the operational safety of the system. Furthermore, a Proportional-Integral (PI) control loop stabilizes the system response under varying load conditions, ensuring consistent output behaviour. Thermal stability is supported by the use of an aluminium heat sink on the output diode, which promotes reliable operation even under continuous load.

The project is fundamentally aimed at achieving a highly efficient and reliable energy conversion process by leveraging a carefully optimized circuit architecture, effective thermal management, and proper control strategies. Rather than targeting specific numerical thresholds, the design prioritizes minimizing losses, improving energy usage, and enhancing overall system performance through thoughtful engineering choices. The compact 13x9.5 cm PCB layout further supports the converter's applicability in space-constrained applications such as consumer electronics, industrial systems, and portable power solutions. Additionally, an integrated overvoltage protection (OVP) circuit strengthens the system's resilience against abnormal voltage conditions, contributing to long-term operational reliability.

In essence, this project highlights a design philosophy centred on sustainability, efficiency, and practicality, aligning with modern demands for affordable, clean, and dependable energy solutions.

Determination of basic parameters for design:

Table I- 2 Basic parameters for design

Circuit Topology	Flyback Converter
Power Level	60W
Output Voltage	20V
Output Current	3A
Input Power Factor	%85
Power Conversion Efficiency	%85
PWM Control IC	UC3844
Protection Circuit	Over Voltage Protection
Printed Circuit Board Size	13x9.5

II. CIRCUIT DESIGN

Reference to subsection 2.1., 2.2., 2.3. Reference to subsection 2.2.1., 2.2.2., 2.2.3., 2.2.4., 2.3.1. Reference to subsection 2.2.1.1., 2.2.1.2, 2.2.1.3., 2.2.1.4., 2.2.1.5., 2.2.1.6, 2.2.2.1., 2.2.2.2., 2.2.4.1. Reference to Figures 2-1, 2-2, 2-3, 2-4, 2-5, 2-6, 2-7, 2-8, 2-9, 2-10. Reference to Tables 2-1. Reference to citations [1], [2], [3], [4], [5], [6], [7].

2.1. Flyback Topology General Features

The flyback converter is one of the most cost-effective and practical types of isolated converters. It is widely used in applications requiring electrical isolation and moderate power levels. However, due to limitations related to transformer design and efficiency, it is typically applied in systems operating at power levels below 100 W.

To better understand the working principle and components of the flyback topology, we can simplify the analysis by focusing on a DC-DC flyback circuit, ignoring the rectification of the AC input voltage for now. This simplification highlights the core concepts of its operation. For this purpose, the circuit containing the basic components of the flyback topology is given in Figure 2.1.[1]

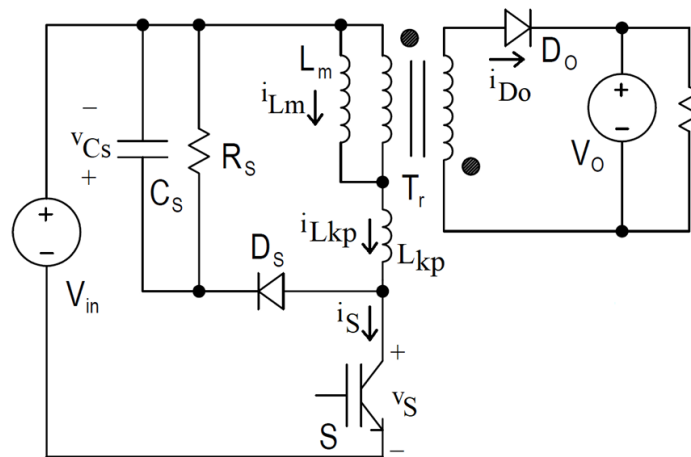


Figure II- 1 Basic circuit diagram of flyback converter [2]

In the flyback converter, unlike other converters, the primary and secondary currents are also magnetizing currents. When the switch is on, no energy is transferred to the output. When the switch is off, the energy stored in the primary is transferred to the output. However, the energy of stray inductances cannot be transferred to the output. Therefore, protection against stray inductance is required. We provide this with snubber.

In Figure 2.1, R_s , C_s and D_s represent the suppression elements of the flyback converter. L_m is the magnetizing inductance, L_{kp} is the leakage inductance. The output capacitor is shown as a voltage source.

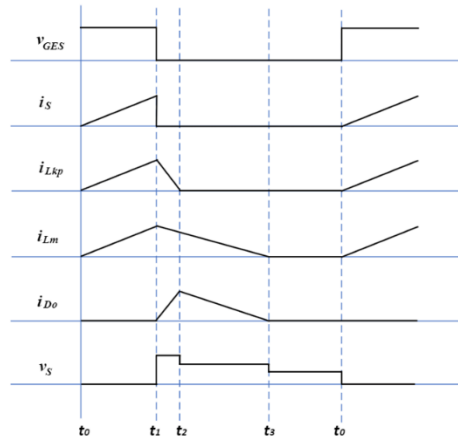


Figure II- 2 Basic waveforms of flyback converter [2]

The waveforms of the flyback in 4 different time intervals are given in Figure 2.2. In the initial state, the signal of the switches is not applied, and the load is fed by the output capacitor. The signal of the switch is applied and the $t_0 - t_1$ interval begins. In this interval, D_o is cut off. The input voltage is applied to the primary. Magnetization and leakage inductance currents increase linearly.

The signal of the switch is cut off and the $t_1 - t_2$ interval begins. In this interval, the equality of the magnetization and leakage inductance currents is disrupted and D_o enters conduction. D_s goes to cut-off and $t_2 - t_3$ interval starts. In this interval, the secondary winding has taken over all the magnetizing current. The magnetizing inductance is reset, the output diode goes to cut-off and this interval ends, returning to the starting interval.

2.2. Power Circuit

2.2.1. Primary Side

We start designing our power circuit from the AC input side, which is the primary side of the transformer. Figure II.3 is taken from the drawing of the primary side on Altium Designer.[3]

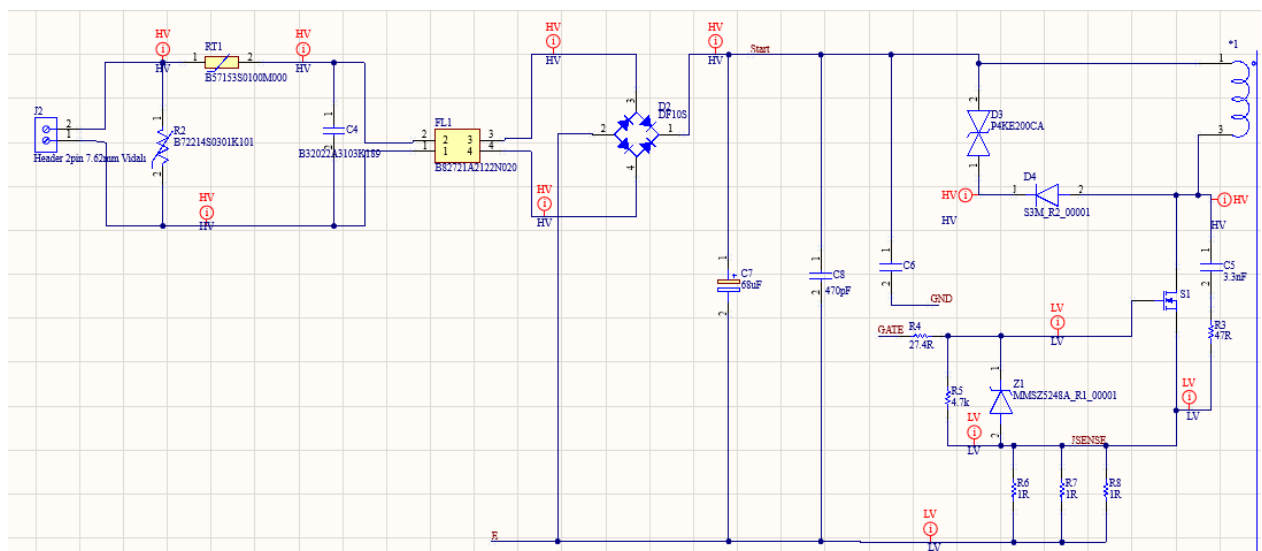


Figure II- 3 Primary side of power circuit

Here it is necessary to make some basic calculations about the circuit.

The first step of the calculations was to determine the minimum and maximum input voltage values. The input voltage was calculated with a 15% tolerance.

$$V_{in} = 230 \pm \%15 \quad \text{Equation 2.1}$$

$$195VAC \leq V_{in} \leq 265VAC \quad \text{Equation 2.2}$$

$$195\sqrt{2} = 276V \quad \text{Equation 2.3}$$

When we take the voltage fluctuation on the capacitor as 15%, Equation 2.4 occurs.

$$\Delta V_{CAP} = \%15 \quad \text{Equation 2.4}$$

$$276 \times 0.85 = 235V$$

$$\frac{276+235}{2} = 235V \quad \text{Equation 2.5}$$

$$I_{CAPDISC} = \frac{60/0.8}{255} = 296 \text{ mA} \quad \text{Equation 2.6}$$

$$\Delta V = 276 - 235 = 41V \quad \text{Equation 2.7}$$

$$\frac{I}{C} \times \Delta t = \Delta V \quad \text{Equation 2.8}$$

$$\frac{296}{C} \times 8.22 \text{ ms} = 41V$$

$$C = 60\mu F$$

Here the input capacitor is found to be 60μF.

When it is desired to calculate the duty cycle, the inductance of the transformer and the current passing through the primary's main loop, the following steps are followed.

$$V_{REFLECTED} = 20 \times 5 = 100 \text{ (Comes in 5 transformer turns ratio)} \quad \text{Equation 2.9}$$

$$235 \times D = 100 \times (1 - D) \quad \text{Equation 2.10}$$

$$D = 0.299$$

$$P_{in} = \frac{P_o}{eff} = \frac{60}{0.8} = 75 \quad \text{Equation 2.11}$$

$$D = \sqrt{\frac{2 \times P_{in} \times L}{T}} \times \frac{1}{V_{in}} \quad \text{Equation 2.11}$$

$$L = 329 \mu H$$

$$\frac{1}{2} \times L \times I_{Lp} \times f = P_{in} \quad \text{Equation 2.12}$$

$$I_{Lp} = 2.135 A_{rms}$$

Since the main current path is completed here, the current passing through the primary winding of the MOSFET, capacitor and transformer must be equal. The calculation of this current is given in Equation 2.13 and Equation 2.14.

$$V_{c(ort)} = 255V$$

$$\frac{V_{c(rms)}}{L} \times D_{mean} \times f = I_{Lp} \quad \text{Equation 2.13}$$

$$I_{s(rms)} = I_{pri(rms)} = I_{c(rms)} \text{ at } 100 \text{ kHz} \quad \text{Equation 2.14}$$

$$I_{s(rms)} = 0.656A$$

2.2.1.1. EMI Filter

Systems with high switching frequencies such as SMPS emit high frequency signals to their surroundings and the network they are connected to. These high frequency signals can affect surrounding devices and cause these devices to malfunction. For this reason, an EMI filter is placed at the input of the circuit we designed.

Systems with high switching frequencies such as SMPS emit high-frequency signals to their surroundings and the network they are connected to. These high-frequency signals can affect surrounding devices and cause these devices to malfunction. For this reason, an EMI filter is placed at the input of the circuit we designed. NTC is used in the EMI filter to prevent high-capacity storage capacitors that are empty at the first moment of operation of the system from drawing high current. The current resistance graph was examined according to the current that will pass through the NTC and the heating of the NTC was calculated according to the resistance that will occur here and an optimization study was carried out. As a result, a price analysis was made and a selection was made.[4]

The varistor is placed for protection against the possibility of the network voltage reaching very high levels. The X2 capacitor is placed to filter differential mode noise. The Y2 capacitor is placed between phase and ground. It is selected according to the voltage (input voltage) that will fall on the X2, Y2 capacitors and the Varistor. The common mode choke is placed between phase-phase to filter common mode noise. The CMC was selected by performing heating analyses according to the current value passing through it.

2.2.1.2. Rectifier Diodes

After the EMI filter, the AC voltage value is converted to DC voltage by the full wave rectifier circuit. The necessary calculations for the selection of these diodes were made and as a result, the DF10S - HT product was decided.

$$P_{in} = \frac{P_o}{\eta_{eff}} = \frac{60}{0.8} = 75 \quad \text{Equation 2.11}$$

$$D = \sqrt{\frac{2 \times P_{in} \times L}{T}} \times \frac{1}{V_{in}} \quad \text{Equation 2.11}$$

$$L = 329 \mu H$$

$$\frac{1}{2} \times L \times I_{Lp} \times f = P_{in} \quad \text{Equation 2.12}$$

$$I_{Lp} = 2.135 Arms$$

2.2.1.3. Snubber

In order to reset the leakage inductance, the snubber voltage must be higher than the reflected voltage. If the snubber voltage is too high, the voltage on the MOSFET increases and the EMC deteriorates. If it is too low, the reset time of the leakage inductance will be extended, causing the snubber power to increase. In principle, the snubber voltage can be selected twice the reflected voltage.[1]

Calculating the snubber power started with finding the leakage inductance. Since a ready-made transformer was not used and the transformer was wound by hand, the loss was approximately 1% and the calculations were made accordingly.

$$L_k = 0,01 \times 329 \times 10^{-6} = 3,29 \mu H \quad \text{Equation 2.13}$$

$$P_{snubber} = \frac{1}{2} \times L_k \times I_{Lk}^2 \times f \quad \text{Equation 2.14}$$

$$P_{snubber} = 0.75 W$$

The power consumed on the snubber was found to be 0.75W and the heating calculations were made accordingly and the diode selections were made accordingly. Here, the junction to ambient values from the diode datasheet are examined and a calculation is made according to the heating values. The junction to ambient value for this component was determined to be 60°C/W. Accordingly, the calculation is given in Equation 2.15.

$$P_{snubber} \times R_{JA} = 0.75 \times 60 = 45^\circ C \quad \text{Equation 2.15}$$

2.2.1.4. Transformer

The AC voltage generated by the switching of MOSFETs is transferred to the load side (secondary) of the circuit by the transformer. The main purpose of the transformer in between is to provide isolation between the load and the power system. Since a current very close to the AC current passes through the transformer, Litz wire is used in the transformer winding to prevent losses due to the skin effect.

The transformer winding ratio is designed as 5:1. The core selection of the SMPS transformer is determined according to the power value. Considering that flyback will be used, the use of ETD34 ferrite core at 60kW power level is deemed appropriate. The effective parameters of the ETD34 core are given in Table2.1.

Parameter	Symbol	Value	Unit
Core constant	C1	0.815	mm ¹
Effective path length	l _e	79.0	mm
Effective area	A _e	97.0	mm ²
Effective volume	V _e	7660	mm ³
Center leg area	A _c	91.6	mm ²
Winding area	A _w	187.0	mm ²
Weight of set	W	39	g

Figure II- 4 Effective parameters of the ETD34 [5]

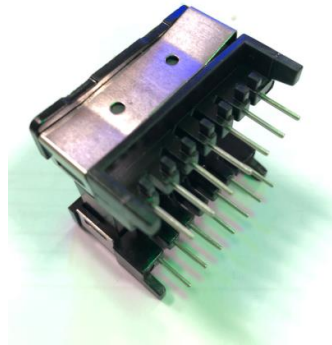


Figure II- 5 ETD34 carcass[6]

Other important parameters that we will use in transformer design are given in Table 2.2.

Table II- 1 Some parameters for transformer design

Primer:Seconder (n)	P _o	I _{pr(rms)}	I _{pr(peak)}	I _{sec(rms)}	I _{sec(peak)}	L _{pr}
5	60W	0.646A	2.135A	5.25A	10.675A	329μH

$$N_{pri} \times \Delta\phi = V \times \Delta t$$

Equation 2.12

$$\phi_{\max} = B \times A_e$$

Equation 2.13

$$N_p \approx 40$$

$$N_{sec} = 8$$

The B (magnetic flux) value is selected between 0.6T and 0.2T. If the magnetic flux value is greater than 0.6T, it can pass leakage currents and if it is lower than 0.2T, it can go into saturation. Considering the temperatures in the environment where this card will be located, it was selected as 0.5T.

$$J = 5 \frac{A}{mm^2}$$

Primary wire cross-sectional area:

$$A_{cu(pri)} = \frac{I_{pri(rms)}}{J} = \frac{0.646}{5} = 0.130 \text{ mm}^2$$

Equation 2.14

Secondary wire cross-sectional area:

$$A_{cu(sec)} = \frac{I_{sec(rms)}}{J} = \frac{5.25}{5} = 1.05 \text{ mm}^2$$

Equation 2.15

$$A_{\%} = \frac{N_{pri} \times A_{cu(pri)} + N_{sec} \times A_{cu(sec)}}{A_w} \times 100 \quad \text{Equation 2.16}$$

The proportion of the frame window area occupied is found to be 7.4% from Equation 2.16.

2.2.1.5. MOSFET

PJMB390N65EC_T0_00601 with 650V withstand voltage, 390mΩ RDS(on) value, TO263-3 (D2PAK-3) case was selected as MOSFET. It has 19nC gate charge value for 10A and 650V Drain – Source voltage. Gate charge value is quite low. Thus, it can be easily driven with UC3844 which has 1A driving peak current value. Conduction loss of MOSFET is calculated as 0.163W from Equation 2.17. MOSFET's conduction loss is calculated as MOSFET output capacitance value is given as Coss = 29pF. Thanks to DCM operation, MOSFET enters conduction with ZCS but parasitic capacitor energy is not recovered.[7] Therefore, turn-on loss is found as 0.36W from Equation 2.18. Turn-off loss can be approximately close to turn-on loss or slightly more. That's why the calculations were made by taking it as 0.5W.

$$(I_{s(rms)})^2 \times R_{ds(on)} = (0.646)^2 \times (0.39) = 0.163W \quad \text{Equation 2.17}$$

$$f_{sw} \times \frac{1}{2} \times C_{oss} \times (V_{in})^2 = (100 \times 10^3) \times \frac{1}{2} \times (29 \times 10^{-12}) \times (500)^2 = 0.364W \quad \text{Equation 2.18}$$

When a heating calculation is made according to the junction to ambient values, it shows that the room temperature has risen above approximately 60 °C . Considering the areas where the card will be used, it is seen that this will not cause a problem. For this reason, considering the costs, it was decided that this mosfet was suitable.

2.2.1.6. Current Measuring Resistors

In normal operation, the primary current peak value is determined as 2.135A and the transformer core maximum magnetic flux density is determined as $B_{max} = 0.2T$. In transient regimes, this value can be allowed to increase up to 0.3T. After 0.3T, the core goes to saturation. UC3844, when the Isense pin voltage exceeds 1V, cuts off the switch for protection purposes. According to these data, the resistance values and packages are calculated with the equations given below.

$$\begin{aligned} V_{sense} = 1V & \rightarrow B_{core} = 0.3T \\ B_{core} = 0.2T & \rightarrow V_{sense} = 0.666V \\ R_{sense} &= \frac{0.666}{2.135} = 0.312\Omega \end{aligned}$$

To obtain this sense resistor, 3 parallel 1Ω's were used.

$$Pr(sense) = (I_{s(rms)})^2 \times R_{sense} = (0.646)^2 \times 0.312 = 0.130W$$

When we examine the power of the packages, the SMD 1206 case is suitable.

2.2.2. Secondary Side

It is the part isolated from the primary side by the transformer. Output values are provided from here. Figure II.6 is taken from the drawing of the primary side on Altium Designer.[3]

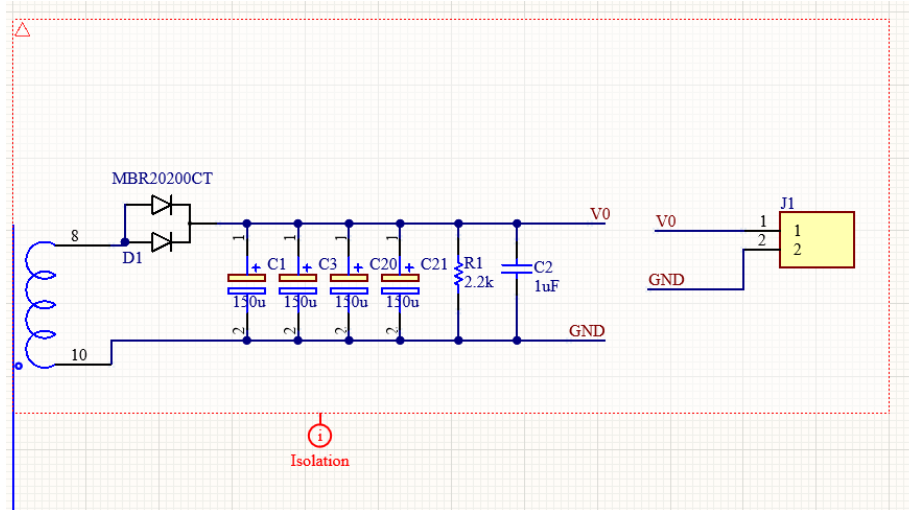


Figure II- 6 Secondary side of power circuit

2.2.2.1. Output Diode

Thanks to the soft switching provided by DCM operation, the output diode switching loss is low. Since soft switching is provided in DCM operation, RC snubber on the output diodes is not essential. In addition, since we work at high frequency, since Schottky diode will be used at the output, switching losses are further reduced and we reduce power loss.

Looking at the datasheet, MBR20200CT has a T0_00001 diode with a forward voltage of 0.52V when 3A passes through it. The voltage and power calculations are given in the equations below.

$$264 \times \sqrt{2} = 374V$$

If we divide this maximum voltage in the primary by the transformer winding ratio, we can find the maximum voltage value that can reach the anode of the diode in the secondary.

$$\frac{374}{5} = 74.8V$$

$$V_d(\text{peak}) = 74.88 + 20 = 94.8V$$

Here the minimum voltage was found. There must be a voltage above this.

$$P_d(\text{loss}) = V_d(\text{forward}) \times I_o = 0.52 \times 3 = 1.06W$$

$$1.06 \times 60 = 63.6^\circ C$$

Since this diode is a THT diode, cooling cannot be provided via the PCB board. Since the heating is high due to power loss, it was deemed appropriate to use it with a heat sink.

2.2.2.2. Output Capacitors

Output capacitors must be selected according to the ripple current that will pass through them. Also, the ESR value must be low enough to provide the desired output voltage ripple. The rms value of the current that will flow through the output capacitor was found to be 3.649A through simulation. Accordingly, 4 identical 50V capacitors with 1A current ripple were used.

SMD ceramic capacitor can also be preferred as output capacitor, however, electrolytic capacitor with higher capacitance was preferred in order to avoid high voltage peak at the output at the first start and to ensure more stable operation of the feedback. Since the powers of other auxiliary outputs are low, capacitors with lower ripple current values were used. In order to reduce the cross-regulation effect and to ensure that the capacitor is discharged after the power is cut off, resistors that will create low power consumption were used at each output (dummy load).[7]

A ceramic capacitor in parallel with the electrolytic capacitor is used to block high frequency noise, i.e. ESL. A ceramic capacitor in parallel with the electrolytic capacitor is used to block

high frequency noise, that is, ESL. An LC filter can also be used to achieve this at the output, but since the cost was tried to be reduced, this method was more economical.

2.2.3. Controller

On the control side, UC3844 was chosen as the PWM integrated circuit because of its cost, easy availability and its previous use in our work. TL431 was used for precise adjustment of the output voltage. PC817 was deemed appropriate to be used as an opto to provide the isolation between them. Figure II.7 is taken from the drawing of the controller side on Altium Designer.[3]

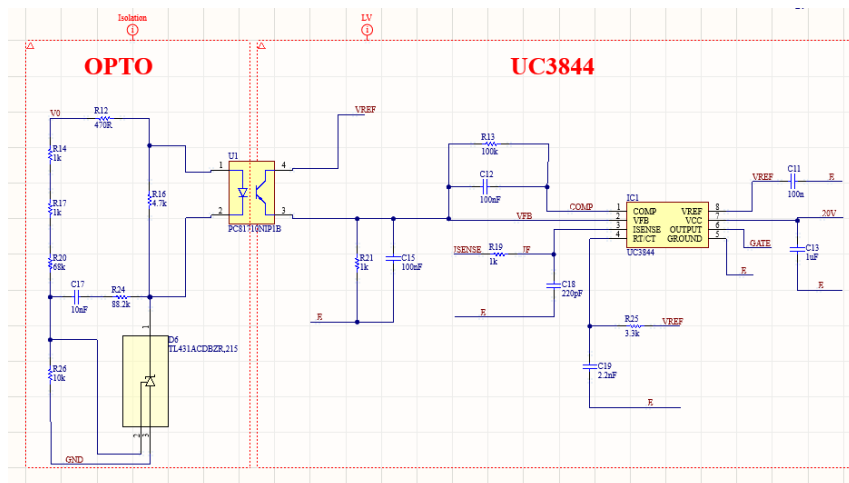


Figure II- 7 Controller side

The feedback part is placed as an estimate based on past experiences and changes will be made during the circuit operation phase.

The reference voltage provided by UC3844 is 5V. With this reference voltage, we make the calculations of our circuit by providing current through R1 and flowing current through the opto. Also, the general purposes of the other components are:

- C17 and R24 are used for stable operation of TL431.
- The reference voltage changes slightly in response to the TL431 current. R16 is used to compensate for this change.
- R21 and R12 affect the feedback gain.
- C15 is used as bypass capacitor to generate 2.5V.
- C18 and R19 ensure that the measured current is filtered and reaches the integrated circuit.
- C19 and R25 are used to set the switching frequency to 100kHz.
- C11 and C13 are bypass capacitors close to the integrated circuit.

2.2.4. Protection Circuit

In general, it was found appropriate to add a protection circuit to the circuit. Details are given under heading 2.2.4.1.

2.2.4.1. Over Voltage Protection (OVP)

Basically, in the circuit where we use the opamp(LM321) as a comparator, it scales the output current and voltage values and compares them with the reference values. The opamp in the circuit is used as a comparator. In cases of high current and high voltage conditions compared to normal current and voltage conditions, a high signal is generated at the opamp's output and a voltage higher than 2.5V is applied to the VFB pin of UC3844, preventing the circuit from operating at high voltage. Figure II.8 is taken from the drawing of the OVP on Altium Designer.

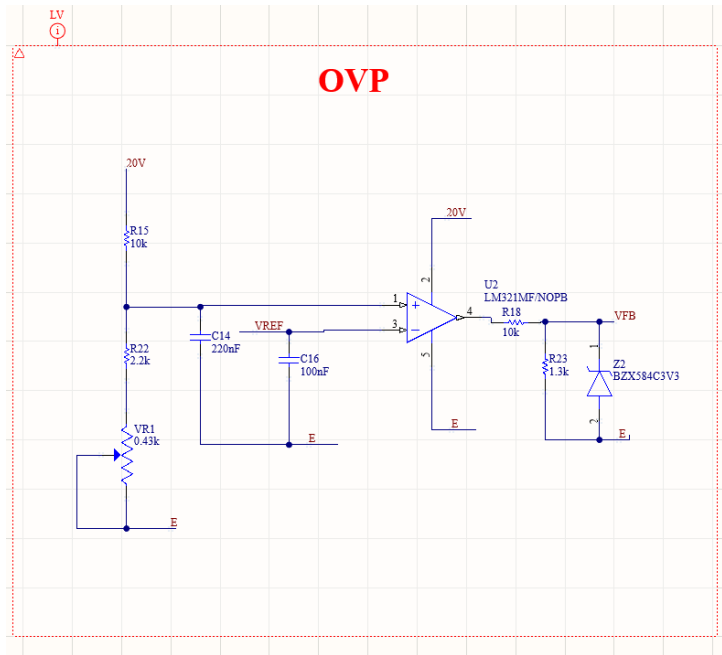


Figure II- 8 Over voltage protection circuit

2.3. PCB Desing

The PCB design was done via Altium Designer.[3] The final design is shown in Figure II-9 and Figure II-10.

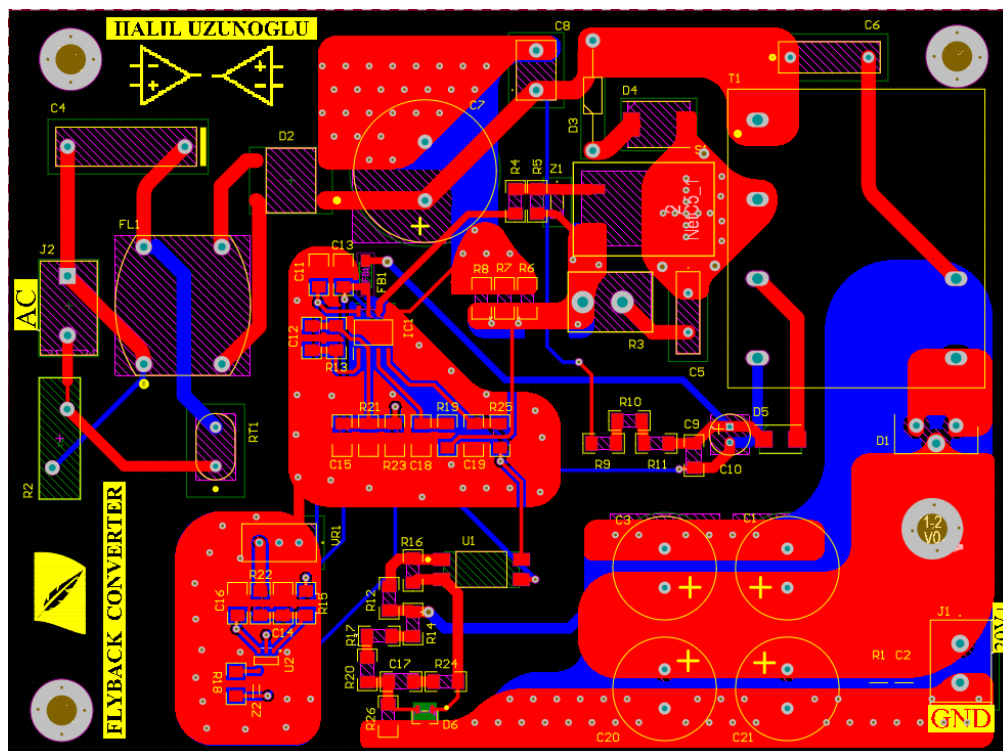


Figure II- 9 Final PCB design top layer

2.3.1. Design stages

The arrangement of components is very important first of all for their operation and then for efficiency. While arranging the components, the current that will pass through the main current path was taken into consideration and the paths were drawn in a way that would handle the current. In order to increase efficiency, incoming and outgoing paths were tried to be drawn as parallel to each other as possible in order to reduce stray inductance.

The source pin of the MOSFET was drawn as large as possible to increase the cooling of the MOSFET.

Since high current will pass on the output side, the paths were drawn as polygons and parallel to each other as much as possible to reduce stray inductance.

In general, efficiency was tried to be increased by trying to make current loops as short as possible.

Figure II-11 shows the 3D view of the PCB, giving a preview of how it will look when completed.

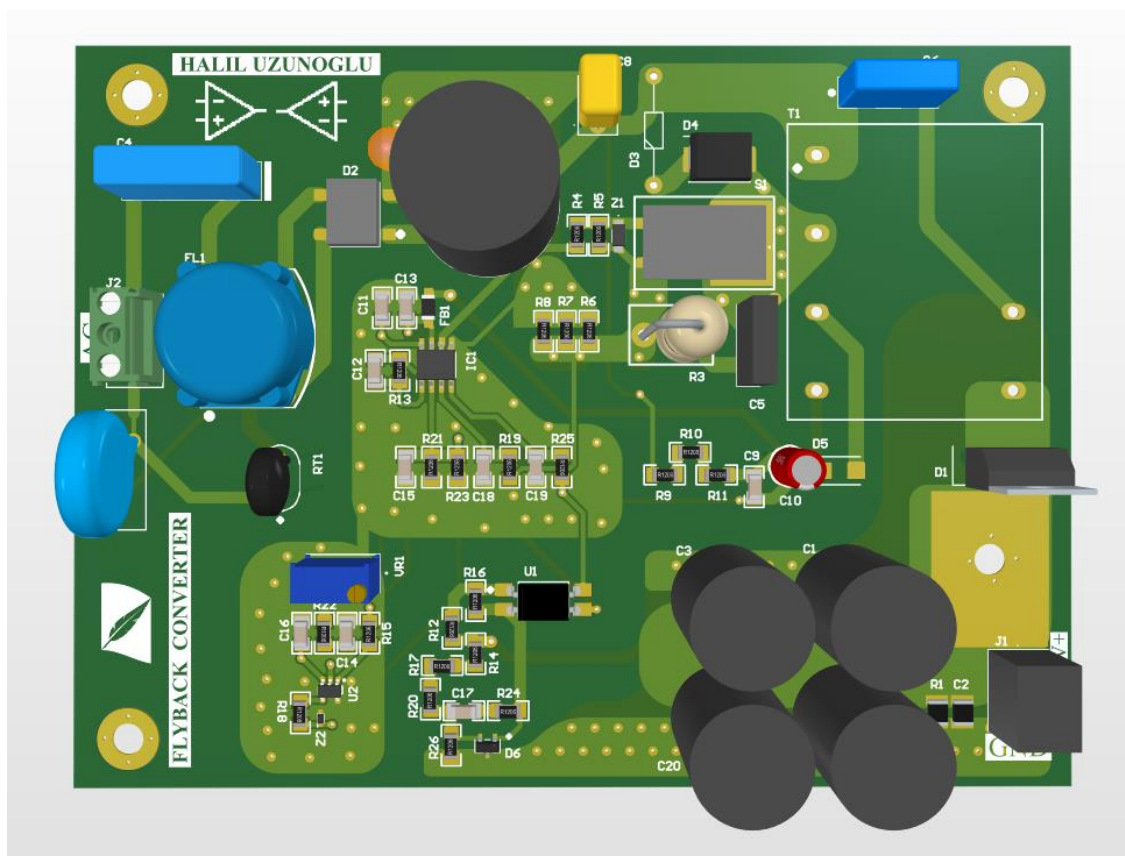


Figure II- 11 3D view of the PCB

III. APPLICATION AND TEST OUTPUTS

Reference to Figures 3-1, 3-2, 3-3, 3-4, 3-5, 3-6, 3-7, 3-8, 3-9, 3-10, 3-11, 3-12. Reference to Tables 3-1.

In this section, the measurement results taken from the converter are given. All measurement results are taken for V_{in} : 220V_{AC}.

The circuit's product on the PCB is shown in Figure III-1.



Figure III- 1 The circuit is implemented on a PCB

In Figure III-2, the output voltage waveform taken from the oscilloscope when the output is unloaded is given. Figure III-3 shows the output voltage waveform taken from the oscilloscope when the output is loaded with 50W.

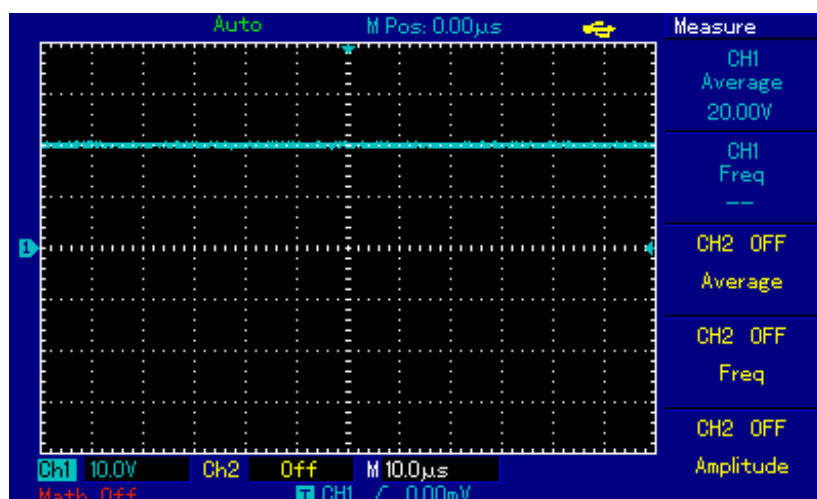


Figure III- 2 No-load output voltage

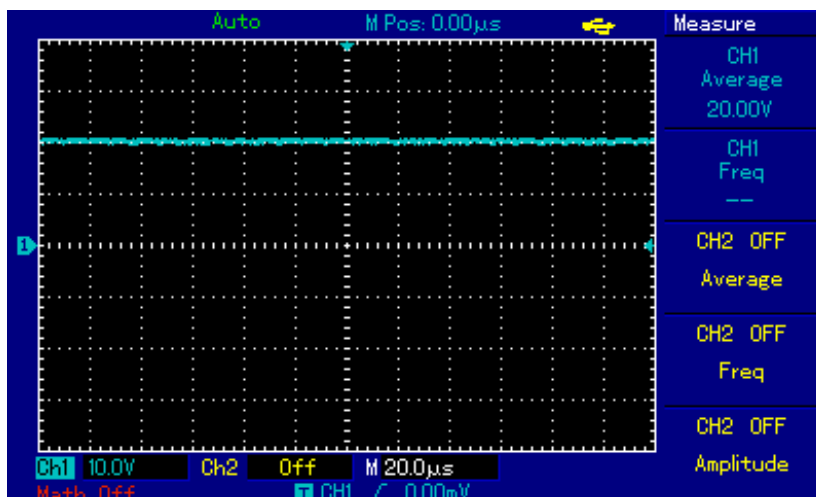


Figure III- 3 Output voltage with 50W load

Figure III-4 shows the MOSFET V_{DS} and V_{GS} voltage waveforms taken when the circuit is loaded with 50W. It is seen that the converter operates in interrupted current mode (DCM) at approximately full load as designed. Figure III-5 shows a close-up of this waveform. This waveform is the waveform that the flyback topology should have. A zoomed-in version of the MOSFET V_{DS} and V_{GS} voltage waveforms is given in Figure III-5.

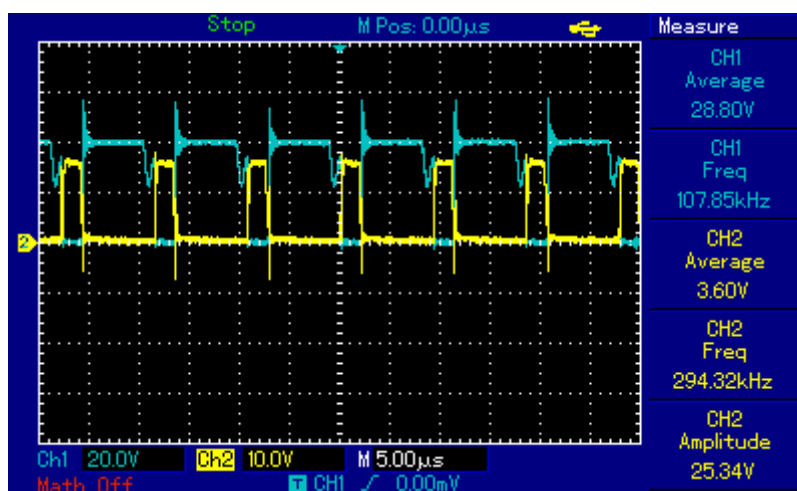


Figure III- 4 MOSFET V_{DS} and V_{GS} voltage waveforms

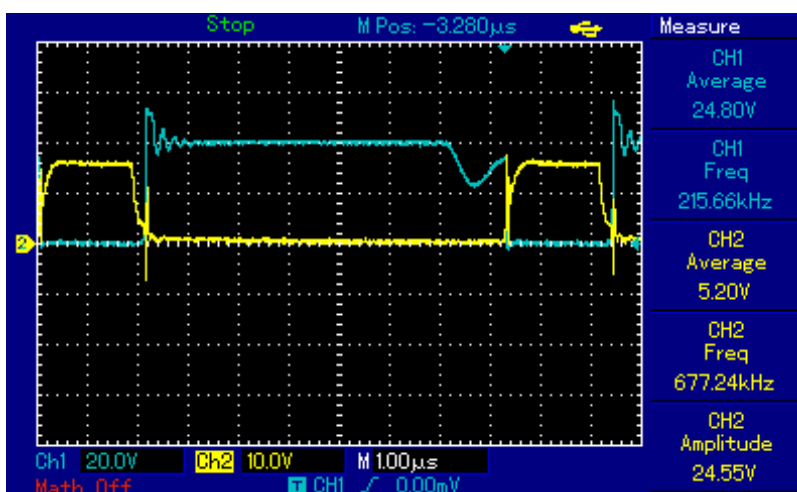


Figure III- 5 Zoomed in on MOSFET V_{DS} and V_{GS} voltage waveforms

Figure III-6 The voltage waveforms on the V_{GS} and sense resistors at 50W load are given. The current measurement was taken directly from the resistor without a filter. In Figure III-7, the scaled and zoomed version of the wave is given.

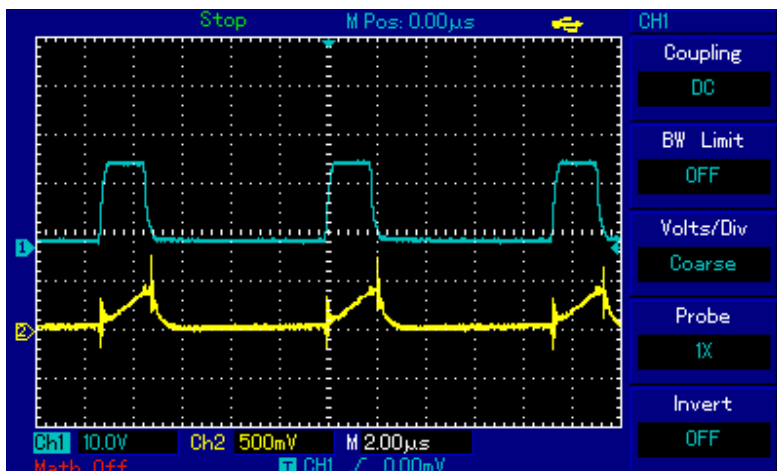


Figure III- 6 Zoomed in on MOSFET VDS and VGS voltage waveforms

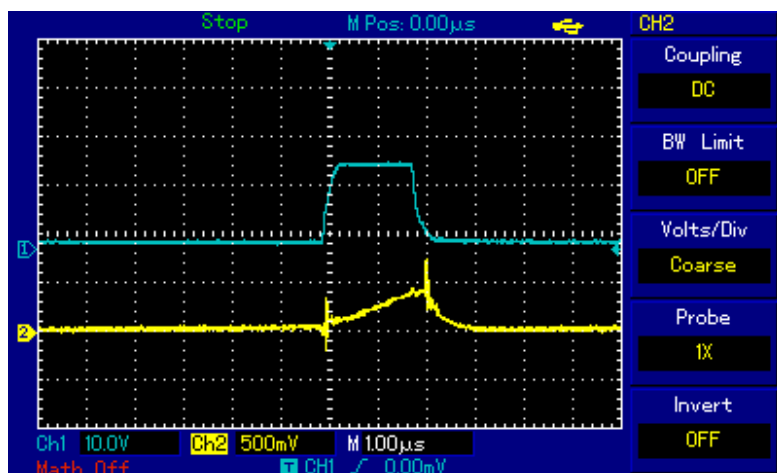


Figure III- 7 Zoomed in voltage waveforms on VGS and sense resistors at 50W load

In Figure III-8, the voltage waveforms taken directly from the 3rd pin of the UC3844 (filtered) for the switch current measured over the VGS and sense resistors at a 50W load are given. In Figure III-9, a zoomed-in version of this waveform is given.

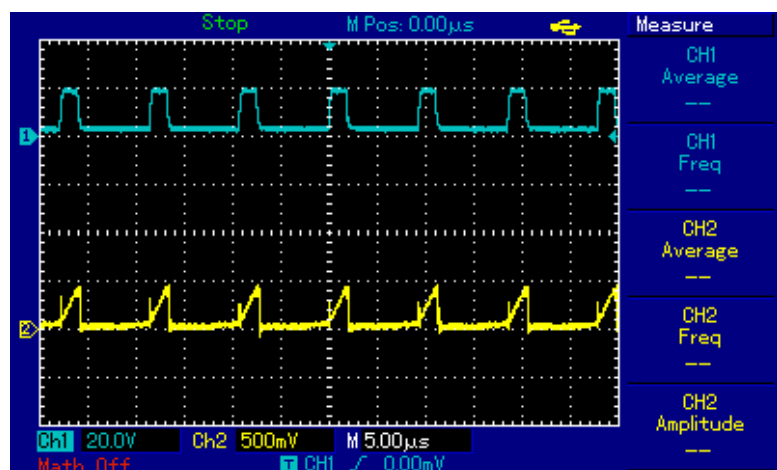


Figure III- 8 Voltage waveforms taken from section 3 of VGS and UC3844 at 50W load condition

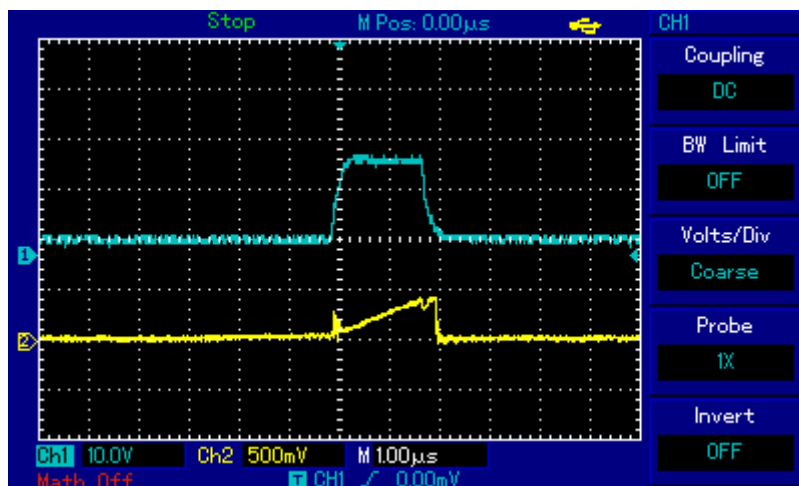


Figure III- 9 Zoomed in voltage waveforms taken from section 3 of VGS and UC3844 at 50W load

The voltage waveform of the output diode in our secondary winding is given in Figure III-10. A zoomed-in version of this voltage waveform is given in Figure III-11.

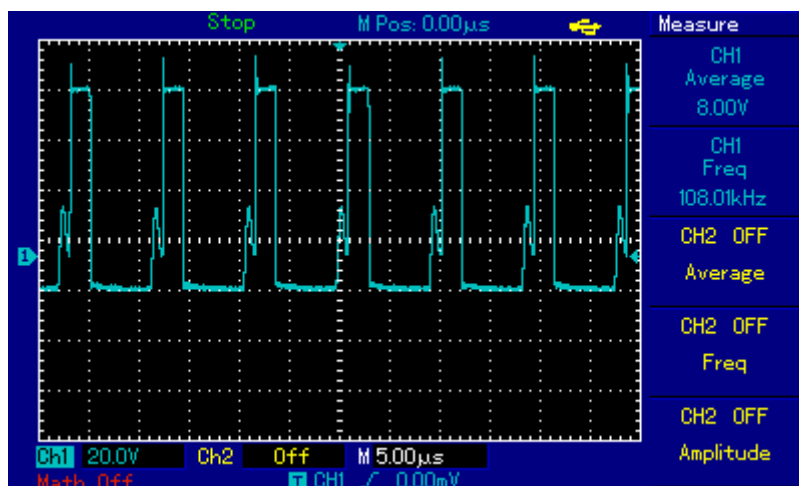


Figure III- 10 Voltage waveform of the output diode in the flyback

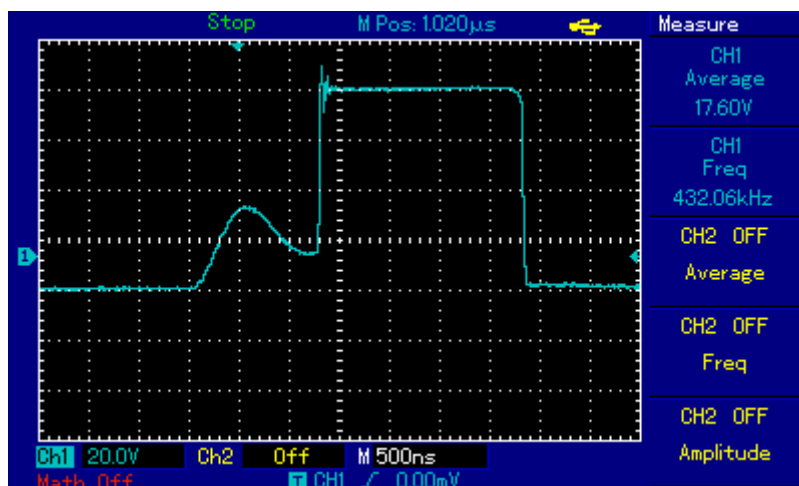


Figure III- 11 Zoomed in on the voltage waveform of the output diode in the flyback.

In Figure III-12, the steady state thermal measurement of the converter taken at room temperature at 50W load is given.



Figure III- 12 Steady state thermal measurement of the converter taken at room temperature under 50W load condition

Table III- 1 Work-Packages Schule

WP Num	Name and Objectives of Work Packages	Time Range
1	Schematic design and presentation development: In this phase, the full schematic diagram of the converter circuit was created using Altium Designer. Component selection, EMI filter, control, and protection circuits were planned, and initial presentations were prepared	October - December
2	PCB design completion and order placement: The PCB layout was designed based on the finalized schematic. High-current paths, ground planes, and component placement were optimized. PCB Gerber files were prepared and sent for manufacturing.	December - February
3	Component assembly and transformer preparation: After receiving the PCB, all components were mounted onto the board. The custom transformer (ETD34 ferrite core) was wound according to design parameters and installed. Visual inspection and continuity tests were performed.	February - March
4	Circuit testing, performance measurements, and final reporting: The assembled board was tested under no-load and full-load conditions. Output voltage, switching waveforms, and thermal performance were measured. Test results were recorded and final documentation was completed.	March - April

IV. RESULT AND DISCUSSION

4.1. Result and Technical Discussion

During the experimental validation phase, the created flyback converter demonstrated stable and trustworthy performance under varying load conditions. During both no-load and full-load assessments, the output voltage remained regulated within the anticipated tolerance range, validating the efficacy of the feedback control loop and transformer design. Oscilloscope assessments taken from the MOSFET VDS and VGS terminals indicated proper high frequency switching characteristics, consistent with anticipated operation in Discontinuous Conduction Mode (DCM). The stability of the waveform further suggested that the Proportional-Integral (PI) control approach effectively reduced transient fluctuations during load changes. PCB layout techniques, including the parallel routing of high-current paths and ground plane enhancement, played a significant role in system stability and low electromagnetic interference (EMI) levels. Measurements indicated minimal noise coupling, and the control circuitry upheld signal integrity across different operational conditions. The compact PCB design enabled efficient current paths and reduced stray inductance, further improving the converter's electrical performance.

In conclusion, the experimental findings confirmed that the converter achieved its intended performance objectives regarding voltage regulation, thermal control, and system stability. The design decisions made during the development stage, such as optimized magnetic components, feedback control methods, and PCB layout considerations, directly helped attain a dependable and efficient energy conversion system. Future enhancements like the addition of input surge protection or advanced switching methods might be investigated in future versions to expand application possibilities and enhance overall system efficiency.

4.2. Restrictions & Conditions:

4.2.1. Economical

In this project, analyses regarding economic feasibility were performed based on availability and cost. All used circuit elements were chosen from inexpensive components that are commonly found in the market. To avoid tax costs, companies that supply domestic components were utilized. The estimated total cost of producing and testing the prototype is around 55 USD. Furthermore, the cost for producing the PCB is 35 USD. During component selection, calculations regarding the lifespan of the products were conducted in accordance with the conditions they operate in the circuit, and the most appropriate one was attempted to be selected based on its price from the datasheet.

The design aimed for 85% energy efficiency, and this value was confirmed through testing. With its high efficiency, the converter significantly decreases energy consumption and delivers substantial energy savings over long-term use. This has a positive impact on the project's ROI (Return on Investment) rate. For instance, it is expected that when used in an industrial system, this converter will utilize 15% less energy annually, leading to a reduction in annual operating costs and a shorter return on investment timeline. Moreover, the compact design of the device, which consists of a limited number of components, decreases the size of the PCB, and these aspects lower mass production costs while providing a competitive price edge in the market. Therefore, an economical solution is presented for both individual users and industrial manufacturers.

4.2.1.1. 7th Goal: Affordable and Clean Energy

This project directly supports Article 7: Affordable and Clean Energy goal established within the United Nations Sustainable Development Goals (SDGs) framework. SDG 7 seeks to enhance access to reliable, sustainable, modern, and affordable energy for all individuals.

The engineered AC-DC flyback converter successfully maximizes energy usage, particularly due to its high-frequency switching, optimal transformer design, and effective control circuit architecture. This architecture prioritizes minimizing power loss and maximizing performance throughout the system, offering a solution that boosts energy efficiency.

The compact design, cost-effectiveness, and ease of manufacturing of the converter created as part of the project aid in making energy access more affordable. Furthermore, due to its minimized power loss and optimized design aspects, it facilitates the long-term reduction of energy usage and carbon emissions.

With these attributes, the project addresses affordable and clean energy objectives not only in individual applications but also in sectors such as renewable energy systems, portable energy solutions, and industrial automation systems. Therefore, it directly contributes both technically and economically to the realization of the SDG 7 objective established on a worldwide level.



Figure IV- 1 7th Goal: Affordable and Clean Energy

4.2.1. Manufacturability

In this project, the design was created by taking manufacturability into account from the start. The circuit schematic design and the printed circuit board (PCB) layout were developed using Altium Designer software. By organizing all components on a 2-layer PCB, particularly on the top layer, both manual soldering was made easier and suitability for automated production was guaranteed.

During the setup of the components, high current paths (notably in the input-output and MOSFET area) were crafted with short and thick paths, which minimized resistance and heat loss, leading to more dependable production. Surface mount (SMD) and through-hole mount (THT) components were combined, yet during assembly, it was assured that all components resided on the same surface, considering production ease.

Details such as the winding ratio, core type (ETD34), and wire diameter needed for transformers and similar specialized components were documented. This ensures that when others wish to replicate the project, they can proceed directly to production without facing any uncertainties. Furthermore, all components utilized in the circuit were chosen from brands that are readily available in the market, thus lowering risks associated with the supply chain for production.

With the component layout that considers voltage resistance, power density, and thermal distribution, the system has been guaranteed to function stably without overheating, and post-production issues have been averted. All these design and engineering choices confirm that the project is not merely a single prototype but also a product designed for large-scale manufacturing.

4.2.1.1. 9th Goal: Industry, Innovation and Infrastructure

This project aligns with the United Nations Sustainable Development Goal 9, which emphasizes the development of resilient infrastructure, the promotion of inclusive and sustainable industrialization, and the encouragement of innovation.

The flyback converter developed in this project presents a scalable and manufacturable solution for reliable energy conversion needs, especially in low-to-medium power applications. Its compact design, efficient energy management, and cost-effective component selection make it highly adaptable for integration into industrial control systems, portable electronics, and renewable energy infrastructure.

By focusing on design optimization, manufacturability, and system reliability, this project contributes to advancing industrial technologies that are both sustainable and widely accessible. The use of common, easily sourced components ensures that the converter can

be mass-produced with minimal supply chain limitations, supporting the creation of robust industrial infrastructure. Additionally, the innovative approaches applied in PCB layout, control strategy, and thermal management exemplify how targeted technological improvements can enhance both system performance and production efficiency.

Overall, this project demonstrates a practical contribution to industrial innovation by providing an efficient, replicable, and scalable power electronics solution aligned with modern infrastructure and manufacturing needs.



Figure IV- 2 9th Goal: Industry, Innovation and Infrastructure

4.2.2. Safety

In this project, safety was regarded as one of the key design factors due to the high input voltage (220V AC) and the isolated nature of the system. Several measures were taken during both circuit design and physical layout to ensure electrical safety. Initially, a transformer was positioned between the primary and secondary circuits to provide complete isolation. This method physically prevented the transmission of high voltage to the output side and ensured user safety. Both the transformer winding configuration and the trace gaps on the PCB (clearance and creepage) were designed according to suitable standards for maintaining the isolation.

An opto-isolator (PC817) was employed in the control and feedback section, allowing control signals to be sent from the secondary to the primary without physical contact. With this arrangement, the control circuit and any equipment that might be close to the user were kept isolated from hazardous voltages.

Components that offer protection against overvoltage and sudden voltage spikes (surge) were incorporated on both the input and output sides of the circuit:

- Varistor: Shields against sudden voltage surges at the input.
- NTC thermistor: Provides a soft start effect to counteract sudden increases in current at the input.

- Snubber circuit: Prolongs the lifespan of the device by limiting abrupt voltage spikes on the MOSFET.

- Overvoltage Protection (OVP) Circuit: Regulates the output voltage with TL431 and opamp (LM321). If the designated threshold value is surpassed, the circuit is interrupted by engaging the UC3844 integrated circuit.

Moreover, the risk of heating is reduced, and the likelihood of fire is mitigated with the utilization of the aluminium heat sink linked to the output diode. Paths designed for high current are crafted with broad traces, power density is managed, and an even distribution of heat throughout the PCB is ensured.

Due to all these measures, the circuit not only operates safely but also guarantees long-term and stable performance. Since such safety measures are crucial, especially in industrial settings, the design was deemed compatible with standards like UL and CE.

4.2.3. Reproducibility

The design of this project has been meticulously documented in detail and in a systematic manner so that it can be replicated by various individuals yielding identical results. All circuit diagrams, PCB layouts, component lists (BOM), simulation outcomes, and test results of the

project have been comprehensively prepared and filed. This documentation guarantees the repeatability of the project not only in an academic capacity but also in an industrial context. Detailed part numbers, brand/model information, and technical specifications are provided for all components utilized in the circuit. In this manner, it is feasible to easily substitute any part with an alternative component should it be unavailable. Furthermore, technical details are supplied for components that necessitate special windings, such as transformers. If desired, these can be manufactured by a company due to the provided technical information or, as demonstrated in this project, can be produced in accordance with the specified parameters using ferrite cores, frames, and wire.

The PCB design was executed using Altium Designer, and project files (schematic, PCB layout, Gerber, drill, etc.) were incorporated into the study. The same board can be conveniently produced by another manufacturer using these files. To facilitate post-production testing, all measurement procedures (input/output voltage, performance under load, MOSFET drive signals, diode waveforms, etc.) have been documented with graphical representations, and oscilloscope images have been included.

4.3. Cost Analysis

This project was developed in accordance with the principles of sustainability, cost optimization, and economic feasibility, and it aims to establish an efficient and cost-effective energy conversion system. Within the framework of the project, initial investment costs, prototype production costs, and the requirements for supporting materials were evaluated in a comprehensive manner.

Project costs were analyzed under three primary categories:

- Prototype Production Costs
- PCB Production Costs
- Laboratory Consumables and Auxiliary Material Costs

In this analysis, Prototype Production Costs encompass the procurement of passive components such as transformers, control integrated circuits (UC3844, TL431), MOSFETs, optoisolators, and capacitors, as well as resistors. The total expenditure for these components was estimated at approximately ₺2000.

PCB Production Costs involve the design, production, solder mask applications, and assembly processes of the double-layer 13x9.5 cm printed circuit board. The total expenditure for PCB production stands at ₺1500, and this amount additionally includes shipping and quality control expenses.

Laboratory Consumables and Auxiliary Material Costs consist of solder wire, flux, cleaning materials, and small hand tools necessary for basic soldering operations. The expenditure for these items was recorded as approximately ₺100.

Through meticulous budget management and resource utilization, the project delivered a cost-effective solution that harmonized high performance with low production costs. Furthermore, the compact and modular design offers the potential for additional reductions in unit costs should mass production occur in the future. Due to the high efficiency of the converter, there was an intention to maintain low operational costs, thereby supporting the return on investment (ROI) and sustainability of the system in the long term.

Table IV- 1 Table Estimated of Estimated Budget Analysis

Cost Item	Estimated Cost (TRY)
Prototype Production	2000
PCB Fabrication	1500
Laboratory Consumables	100
Total Project Cost	3600

4.4. Expected Outcomes

This project successfully achieved the design, implementation, and experimental validation of a 60W AC-DC Flyback Converter, meeting both the technical and performance expectations established at the beginning of the study. Throughout the testing phase, the converter demonstrated stable and efficient operation across different load conditions, consistently delivering a regulated 20V DC output. The use of a TL431-based feedback mechanism, together with an optically isolated control loop and Proportional-Integral (PI) compensation, ensured precise voltage regulation with minimal ripple, even under sudden load changes or fluctuations in input voltage.

The converter operated reliably in Discontinuous Conduction Mode (DCM), as verified through oscilloscope measurements of the MOSFET's VDS and VGS waveforms. This switching strategy not only reduced switching losses but also contributed to soft switching of the output diode, minimizing electromagnetic interference (EMI) and overall thermal stress on the components. The thermal management approach, particularly the aluminium heatsink attached to the output rectifier diode, proved effective in maintaining safe operational temperatures under full-load conditions, preventing thermal runaway and ensuring the long-term reliability of the system.

In terms of layout, the carefully designed double-layer 13x9.5 cm PCB provided a compact and practical solution for medium-power applications. The parallel routing of high-current paths, optimized ground plane connections, and strategic component placement minimized stray inductance and noise coupling, improving both electrical performance and electromagnetic compatibility. Additionally, the integrated Over Voltage Protection (OVP) circuit successfully safeguarded the converter and connected loads against abnormal voltage conditions, enhancing operational safety.

Overall, the project outcomes validated that the proposed flyback converter met its intended objectives in terms of efficiency, voltage stability, thermal reliability, and protection. Moreover, by utilizing widely available, cost-effective components and emphasizing manufacturability, the design offers a scalable, replicable, and sustainable energy conversion solution. Its compact, efficient, and reliable structure makes it highly suitable for applications such as industrial automation, consumer electronics, and portable power systems, while also supporting modern sustainable energy goals through improved efficiency and reduced power losses.

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A. APPENDIX

A.1. Relevant Standards

This project has been designed and implemented in alignment with internationally recognized safety, electromagnetic compatibility (EMC), and power electronics standards. The following global standards are relevant to the design, operation, and potential mass production of the developed AC-DC flyback converter.

4.2.2. IEC 60950-1 - Safety of Information Technology Equipment

This standard specifies the general safety requirements for electrical equipment, particularly focusing on insulation distances, dielectric strength, and protective measures to prevent electric shock hazards. In this project, the isolation between the primary and secondary circuits, PCB trace spacing, and insulation parameters have been designed in accordance with the guidelines outlined in this standard.

4.2.3. IEC 62368-1 — Audio/Video, Information and Communication Technology Equipment Safety

Replacing IEC 60950-1 in many regions, this risk-based safety standard defines requirements based on energy sources and their potential hazards. The design of this converter, especially in terms of thermal protection, surge immunity, and isolation between high-voltage and low-voltage sections, aligns with the requirements of IEC 62368-1.

4.2.4. EN 55032 / CISPR 32 — Electromagnetic Compatibility for Multimedia Equipment

This standard sets limits for radiated and conducted electromagnetic emissions from electronic equipment. Considering that flyback converters typically generate significant high-frequency switching noise, EMI filter design and PCB layout techniques in this project have been developed in accordance with the emission control requirements specified in EN 55032.

4.2.5. EN 61000-4-5 — Surge Immunity Test

This standard defines test methods and performance criteria for evaluating equipment immunity to high-energy voltage transients, such as those caused by lightning strikes or switching surges on the AC mains. In this project, protection components including a varistor, NTC thermistor, and snubber circuits have been incorporated to provide compliance with surge protection guidelines.

4.2.6. IEC 61558-1 — Safety of Power Transformers, Power Supplies, and Similar Products

This standard addresses the general requirements for the design and construction of transformers and power supplies, particularly regarding insulation coordination, temperature limits, and short-circuit protection. The custom-designed transformer in this project, featuring a 5:1 turns ratio and ETD34 ferrite core, has been specified to meet the requirements of IEC 61558-1.

4.2.7. IEC 60068-2 — Environmental Testing for Electrical and Electronic Equipment

This standard series specifies environmental test procedures, including thermal, humidity, and vibration tests, to evaluate the reliability and durability of electronic equipment under operating and storage conditions. The component selection, thermal management, and mechanical robustness of the converter design have been planned with these environmental considerations in mind.

4.2.8. RoHS Directive (2011/65/EU)

The Restriction of Hazardous Substances (RoHS) directive restricts the use of specific hazardous materials in the manufacture of various types of electronic and electrical equipment. Components utilized in this project have been selected with RoHS compliance wherever possible, ensuring environmental safety and material sustainability.

LIST OF THE RELEVANCE OF THE CONSTRAINTS TO THE PROJECT

(The relevancy should be same with the given constraints by the instructor by the beginning of the semester)

(table should be filed by the student and checked by the instructor)

No	Constraint	Yes	No
1	economical,	✓	
2	environmental,		✓
3	sustainability,		✓
4	manufacturability,	✓	
5	ethical,		✓
6	health,		✓
7	safety,	✓	
8	social,		✓
9	legal,		✓
10	political,		✓
11	reproducibility	✓	
12	project cost and a budget analysis.	✓	
13	and, if any, others (as outlined in the 'Thesis Requirement')		✓

Name of the student and signature
Halil Uzunoğlu

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Name of the instructor and signature
Asst. Prof. Erhan DEMİROK

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